Artificial bounding and the subproblem approach to dual Phase 1 coincide iteration by iteration

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Abstract

The artificial bounding method and the subproblem approach are two established ways of starting the dual simplex method when some variables have infinite bounds. The former introduces a slack variable fixed at zero in every equality constraint, takes these slacks as the basis, and exploits the fact that this basis becomes dual feasible at no cost once every nonbasic variable with finite bounds is placed at the bound indicated by the sign of its reduced cost: it replaces infinite bounds by $\mp M$ and applies the dual simplex method, and is known to suffer from the choice of M; the latter minimizes the sum of dual infeasibilities over an auxiliary problem. We show that, for every sufficiently large M, the artificial bounding method generates exactly the same sequence of bases as an M-free two-phase procedure: the dual simplex method on the auxiliary problem of the subproblem approach, followed by the dual simplex method on an *intercept problem* determined by the optimal solution of the auxiliary problem. The two phases are the coefficient of M and the constant term of the basic solution. The result holds for the textbook and the bound flipping ratio tests and for leaving-row rules such as dual steepest edge. Consequently, the two Phase 1 methods differ only in the placement of nonbasic variables with zero reduced cost, no value of M has to be chosen, and the intercept phase distinguishes infeasibility from unboundedness.

Keywords: linear optimization, dual simplex method, dual phase 1, artificial bounds, big-M method, infeasibility and unboundedness

1. Introduction

The dual simplex method needs a dual feasible starting basis. For a basis B with reduced costs $\bar{c}_j$, dual feasibility is obtained by placing every nonbasic variable at its lower bound if $\bar{c}_j \geq 0$ and at its upper bound if $\bar{c}_j \leq 0$, provided that bound is finite. Variables whose required bound is infinite are the only obstacle, and several *dual Phase 1* methods have been developed to remove it; see Maros [13], Koberstein [9] and Koberstein and Suhl [10] for surveys and computational comparisons. Two of them are the subject of this note.

- The *artificial bounding method* replaces every infinite bound by $-M$ or $+M$ for a large constant M and runs the dual simplex method on the resulting bounded problem. It is the big-M method applied to the dual problem [18, 9]. Its weakness is M itself: too small a value cuts off unbounded directions and forces restarts with a larger M, and too large a value causes numerical difficulties. For these reasons Koberstein excluded it from his implementation [9].
- The *subproblem approach* [7, 9, 10] solves an auxiliary problem, obtained by setting the right-hand side and the finite bounds to 0 and the infinite bounds to ∓ 1 (so that variables with two finite bounds stay in it with bounds $[0, 0]$; Koberstein [9] considers the problem obtained from it by removing these variables), whose optimal value is minus the minimal sum of dual infeasibilities. If this value is 0, the final basis is turned into a dual feasible basis of the original problem and the dual simplex method continues; otherwise the problem is reported dual infeasible, which does not distinguish infeasibility from unboundedness.

These two methods are presented in the literature as different techniques. On the primal side it is classical that the big-M method and the two-phase method agree in the limit $M \to \infty$ [2, 1], and treating M as a non-Archimedean infinity turns the big-M method into lexicographic optimization [17, 3]. These statements concern objective values or optimal solutions; they do not say which bases the simplex method visits.

The main result of this note (Theorem 2) is an iteration-level statement on the dual side. For every sufficiently large M, the artificial bounding method visits exactly the bases visited by the following procedure, which involves no M:

Phase A: the dual simplex method on the auxiliary problem of the subproblem approach (the *slope problem*);

Phase B: the dual simplex method, started from the final basis of Phase A, on an *intercept problem* whose bounds are determined by the optimal solution of Phase A.

The proof rests on the fact that, for a fixed basis, the basic solution of the bounded problem is $x^0 + Mx^1$, where x^1 and x^0

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are the basic solutions of the slope and intercept problems. The leaving-row choices of the artificial bounding method first depend only on x^1 and, once x^1 is optimal for the slope problem, only on x^0. The same holds for the decisions of the bound flipping ratio test, whose bound widths are also affine in M, so the result covers the standard implementation techniques of the dual simplex method.

Three consequences follow (Section 4). First, the artificial bounding method and the subproblem approach coincide up to the placement of nonbasic variables with zero reduced cost at the end of Phase 1 (Corollary 5). Second, the behavior of the artificial bounding method for all sufficiently large M is reproduced without choosing M, which removes the weakness noted by Koberstein. Third, the artificial bounding method does not stop when the problem is dual infeasible, and Phase B then decides between infeasibility and unboundedness using dual simplex iterations only (Corollary 3); when a finite optimum exists, nonbasic variables left at artificial bounds are removed by a lexicographic primal push in at most as many pivots as there are such variables (Proposition 6). On the dual side, the theorem is an iteration-level version of the equivalence of the big-M and two-phase methods (Remark 8). We are not aware of a statement of this equivalence in the literature.

2. Setting

We consider

$$\min\ c^\top x \quad \text{s.t.} \quad Ax = b,\ \ l \le x \le u, \tag{1}$$

with $A \in \mathbb{R}^{m\times n}$ of rank m, $l_j \in \mathbb{R} \cup \{-\infty\}$, $u_j \in \mathbb{R} \cup \{+\infty\}$ and $l_j \le u_j$. Inequality rows are assumed to carry slack columns, and equality rows may carry slack columns fixed at $[0, 0]$, so that an all-slack starting basis is available. Let F denote the feasible region of (1). A variable is *fixed* if $l_j = u_j$ and *free* if $l_j = -\infty$ and $u_j = +\infty$.

A family of problems.. All problems below share A and c with (1) and differ only in the *data* $D = (\beta, \lambda, \mu)$, a right-hand side β and bounds $\lambda \le \mu$; we write $\mathrm{P}(D)$ for $\min\{c^\top x : Ax = \beta,\ \lambda \le x \le \mu\}$. We use three kinds of data. Define

$$l^1_j := \begin{cases} -1 & (l_j = -\infty) \\ 0 & (l_j > -\infty) \end{cases}, \quad u^1_j := \begin{cases} 1 & (u_j = +\infty) \\ 0 & (u_j < +\infty) \end{cases},$$

$$l^0_j := \begin{cases} 0 & (l_j = -\infty) \\ l_j & (l_j > -\infty) \end{cases}, \quad u^0_j := \begin{cases} 0 & (u_j = +\infty) \\ u_j & (u_j < +\infty) \end{cases}.$$

- The *bounded problem* $\mathrm{P}(D_M)$, $D_M := (b,\ l^0 + Ml^1,\ u^0 + Mu^1)$ for $M > 0$, replaces every infinite bound of (1) by $\mp M$; the artificial bounding method is the dual simplex method on $\mathrm{P}(D_M)$.

- The *slope problem* $\mathrm{P}(D^1)$, $D^1 := (0, l^1, u^1)$, is the auxiliary problem of the subproblem approach [7]: right-hand side and finite bounds set to 0, infinite bounds set to ∓ 1. Its feasible region is $C \cap [-1, 1]^n$, where $C := \{d : Ad = 0,\ d_j \ge 0 \text{ if } l_j > -\infty,\ d_j \le 0 \text{ if } u_j < +\infty\}$; when $F \ne \emptyset$, C is the recession cone of F. In particular $\mathrm{P}(D^1)$ is feasible ($d = 0$) and has a finite optimal value $z^1 \le 0$.

- For a feasible solution ξ of the slope problem, the *intercept problem* $\mathrm{P}(D^\xi)$, $D^\xi := (b, \lambda^\xi, \mu^\xi)$, has the bounds

$$\lambda^\xi_j := \begin{cases} l^0_j & (\xi_j = l^1_j) \\ -\infty & (\xi_j > l^1_j) \end{cases}, \qquad \mu^\xi_j := \begin{cases} u^0_j & (\xi_j = u^1_j) \\ +\infty & (\xi_j < u^1_j) \end{cases}. \tag{2}$$

  For $\xi = 0$ we have $D^\xi = (b, l, u)$, i.e., the intercept problem is (1) itself.

Explicitly, the three problems are

$$\begin{aligned} \mathrm{P}(D_M):&\ \min\ c^\top x \ \text{ s.t. }\ Ax = b,\ \ l^0 + Ml^1 \le x \le u^0 + Mu^1, \\ \mathrm{P}(D^1):&\ \min\ c^\top x \ \text{ s.t. }\ Ax = 0,\ \ l^1 \le x \le u^1, \\ \mathrm{P}(D^\xi):&\ \min\ c^\top x \ \text{ s.t. }\ Ax = b,\ \ \lambda^\xi \le x \le \mu^\xi. \end{aligned}$$

Note that $D_M = (b, l^0, u^0) + M\,D^1$. A variable is fixed in $\mathrm{P}(D_M)$ and in $\mathrm{P}(D^\xi)$ exactly when it is fixed in (1); in the slope problem every variable with two finite bounds becomes fixed at 0.

States and basic solutions.. A *state* is a pair (B, η) of a basis B (A_B nonsingular) and a status $\eta(j) \in \{L, U, Z\}$ for each $j \in N$, the nonbasic set. Status Z is allowed only for variables that are free in (1) and places them at 0 (cf. [13]). The state is *admissible* for D if λ_j is finite whenever $\eta(j) = L$ and μ_j is finite whenever $\eta(j) = U$. Its basic solution $x(B, \eta; D)$ is defined by $x_j = \lambda_j, \mu_j, 0$ for $\eta(j) = L, U, Z$ and $x_B = A_B^{-1}(\beta - A_N x_N)$. Every state is admissible for D_M and D^1 since all their bounds are finite. With $\bar c_j := c_j - c_B^\top A_B^{-1} A_j$, the state is *dual feasible* if $\bar c_j \ge 0$ for $\eta(j) = L$, $\bar c_j \le 0$ for $\eta(j) = U$ and $\bar c_j = 0$ for $\eta(j) = Z$. This condition does not involve D.

Lemma 1. *For every state (B, η) and $M > 0$, $x(B, \eta; D_M) = x^0 + Mx^1$ with $x^1 := x(B, \eta; D^1)$ and $x^0 := x(B, \eta; (b, l^0, u^0))$.*

Proof. For fixed (B, η), the map $D \mapsto x(B, \eta; D)$ is linear: nonbasic entries copy entries of λ or μ (or are 0), and x_B is linear in (β, x_N). Apply it to $D_M = (b, l^0, u^0) + MD^1$. □

The dual simplex iteration.. Let (B, η) be dual feasible and admissible for D, let $x = x(B, \eta; D)$, and let $\beta(i)$ be the basic variable of row i. The violations of row i are $\delta^-_i := \lambda_{\beta(i)} - x_{\beta(i)}$ and $\delta^+_i := x_{\beta(i)} - \mu_{\beta(i)}$ (equal to $-\infty$ on an infinite side), and $\delta_i := \max(\delta^-_i, \delta^+_i)$. One iteration on $\mathrm{P}(D)$ is as follows.

(a) If $V := \{i : \delta_i > 0\}$ is empty, stop: the state is optimal for $\mathrm{P}(D)$.

(b) *Leaving row.* Choose $r \in \arg\max_{i\in V} \delta_i/\omega_i$, ties broken by a rule τ_{row}. Set $\rho := +1$ if $\delta_r = \delta^-_r$ and $\rho := -1$ otherwise.

(c) *Ratio test.* Let $\alpha_j := (A_B^{-1}A_j)_r$ for $j \in N$ and $E := \{j \in N : \rho\alpha_j < 0\ (\eta(j) = L),\ \rho\alpha_j > 0\ (\eta(j) = U),\ \alpha_j \ne 0\ (\eta(j) = Z)\}$. If $E = \emptyset$, stop: $\mathrm{P}(D)$ is infeasible. Otherwise choose $q \in \arg\min_{j\in E} |\bar c_j|/|\alpha_j|$, ties broken by a rule τ_{col}.

(c′) *Bound flipping ratio test* (alternative to (c)) [11, 14, 9]. Scan E in increasing order of $|\bar{c}_j|/|\alpha_j|$ (ties by τ_{col}), starting with $s := \delta_r$. A candidate j with $\eta(j) \in \{L, U\}$ and both λ_j, μ_j finite is *passed* if $s - |\alpha_j| w_j > 0$, where $w_j := \mu_j - \lambda_j$; it is then flipped to its other bound and s is decreased by $|\alpha_j| w_j$. The first candidate that is not passed is the entering variable q. If $E = \emptyset$ or all candidates are passed, stop: P(D) is infeasible.

(d) *Pivot.* Replace $\beta(r)$ by q in B; $\beta(r)$ becomes nonbasic with status L if $\rho = +1$ and U if $\rho = -1$. With (c′), the passed variables keep their flipped statuses.

We make the following assumptions on the rules.

- The weights $\omega_i > 0$ depend only on A and on the history of states and pivots, not on D. This covers Dantzig's rule ($\omega_i = 1$), the dual steepest-edge rule ($\omega_i = \|e_i^\top A_B^{-1}\|_2$) [6] and dual Devex weights [8].
- τ_{row} and τ_{col} are deterministic and depend on the same information only (for example, smallest index, or the lexicographic rule on perturbed costs for τ_{col} [4]).

Fixed variables are not excluded from E; this keeps the ratio test literally the same for all problems below. The iteration preserves dual feasibility and admissibility, and no new status Z is created. A variable in status Z is never passed in (c′). The same choice of (c) or (c′) is used in all problems below.

3. The main result

For a state that is not optimal for the slope problem, we break ties in the leaving-row selection on P(D^1) by the intercept. The *intercept-refined* rule chooses, among the rows attaining $\max_{i \in V} \delta_i^1/\omega_i$ (where δ^1 denotes the violations in P(D^1)), one maximizing δ_i^0/ω_i, where δ_i^0 is the violation of the same side of row i for the data (b, l^0, u^0); remaining ties are broken by τ_{row}. In other words, the violations $\delta_i^0 + M\delta_i^1$ in P(D_M), divided by ω_i, are compared lexicographically by slope and then by constant term. With (c′), the rule also covers the passing test: writing s^0 for the remaining violation computed in the same way for the data (b, l^0, u^0) (starting from δ_r^0 and using the widths $w_j^0 := u_j^0 - l_j^0$), a candidate is passed if $s - |\alpha_j| w_j > 0$ in P(D^1), or if $s - |\alpha_j| w_j = 0$ and $s^0 - |\alpha_j| w_j^0 > 0$. These are legitimate tie-breaking rules for the dual simplex method on the slope problem: a pass with remaining violation 0 keeps the dual objective unchanged. They are irrelevant whenever no tie occurs.

Theorem 2. *Let (B_0, η_0) be a dual feasible state, and fix rules $\omega, \tau_{\text{row}}, \tau_{\text{col}}$ as above. Consider the following procedure.*

Phase A. *Run the dual simplex method on the slope problem* P(D^1) *from (B_0, η_0) with the intercept-refined rule. Let (B^*, η^*) be the final state, $\xi := x(B^*, \eta^*; D^1)$ and $z^1 := c^\top \xi$.*

Phase B. *Run the dual simplex method on the intercept problem* P(D^ξ) *from (B^*, η^*).*

Then Phase A never stops with infeasibility, ξ is an optimal solution of the slope problem, z^1 is its optimal value, and (B^*, η^*) is a dual feasible state admissible for D^ξ. Assume that both phases terminate. Then there exists $M^\dagger > 0$ such that, for every $M \geq M^\dagger$, the dual simplex method on P(D_M) *from (B_0, η_0) with the rules $\omega, \tau_{\text{row}}, \tau_{\text{col}}$*

(i) visits exactly the states of Phase A followed by those of Phase B, with the same pivots;

(ii) stops in the same way as Phase B (optimal or infeasible);

(iii) has, at every visited state, the basic solution $x^0 + Mx^1$, where $x^1 = \xi$ throughout Phase B and x^0 is the basic solution of the intercept problem in Phase B.

Proof. Phase A and the transition. Phase A is a dual simplex method from a dual feasible state, all bounds of D^1 are finite, and the slope problem is feasible; hence the infeasibility stop cannot occur. At termination, the final state is primal and dual feasible for P(D^1), so ξ is optimal. Call a state *compatible with* ξ if every nonbasic j satisfies $\xi_j = l_j^1$ if $\eta(j) = L$, $\xi_j = u_j^1$ if $\eta(j) = U$, and $\xi_j = 0$ if $\eta(j) = Z$. The final state (B^*, η^*) is compatible with ξ, because the nonbasic entries of its basic solution ξ of the slope problem are l_j^1, u_j^1 or 0 according to the status. Dual feasibility does not depend on the data.

Compatible states. Let (B, η) be compatible with ξ. First, (B, η) is admissible for D^ξ by (2); in particular, so is (B^*, η^*). Second, $x(B, \eta; D^1) = \xi$: this basic solution has the nonbasic entries l_j^1, u_j^1 or 0, which equal ξ_N, and it is the unique solution of $Ax = 0$ with these nonbasic entries, as is ξ. Third, $x(B, \eta; (b, l^0, u^0)) = x(B, \eta; D^\xi)$: the nonbasic entries of the two basic solutions are l_j^0 and λ_j^ξ (status L), u_j^0 and μ_j^ξ (status U), and 0 (status Z), (2) gives $\lambda_j^\xi = l_j^0$ and $\mu_j^\xi = u_j^0$ on these indices, and both right-hand sides are b. Thus, at a compatible state, the basic solution of P(D_M) has slope ξ and its constant term is the basic solution of the intercept problem.

Invariant of Phase B. Every state visited in Phase B is compatible with ξ. By (2), the lower bound of j in P(D^ξ) is finite if and only if $\xi_j = l_j^1$, and likewise for the upper bound. The state (B^*, η^*) is compatible with ξ. In a pivot, the entering variable leaves the nonbasic set, and the leaving variable $\beta(r)$ is placed with status L or U at the finite bound it violated, so it satisfies the condition by this equivalence. A variable flipped in (c′) has two finite bounds in P(D^ξ), so $\xi_j = l_j^1 = u_j^1$ and it satisfies the condition after the flip. Other statuses do not change.

Choice of $M^\dagger$. Let $T_0, \ldots, T_K$ be the states visited by Phases A and B. At each T_k and for each row i, Lemma 1 shows that the violations in P(D_M) are affine in M: $\delta_i^\pm(M) = \delta_i^{\pm,0} + M\delta_i^{\pm,1}$, where $\delta_i^{\pm,1}$ are the violations in P(D^1) and $\delta_i^{\pm,0}$ those for (b, l^0, u^0) (an infinite side of (1) gives $\delta_i^{-,1} = -1 - x_{\beta(i)}^1$, a finite one $\delta_i^{-,1} = -x_{\beta(i)}^1$, and similarly for +). For two affine functions f, h of M, write $f \succ h$ if (f^1, f^0) is lexicographically larger than (h^1, h^0); then $f(M) > h(M)$ for all large M. Choose $M^\dagger$ larger than all finite $|l_j|, |u_j|$, so that D_M has lower bounds not exceeding upper bounds, and large enough that, at every T_k, all numerical comparisons among the finitely many affine functions $\delta_i^\pm(M)$, $\delta_i^\pm(M)/\omega_i$, the remaining violations in (c′) and 0

agree with $\succ$. This is possible since K is finite and the weights ω_i at T_k are fixed numbers determined by the history (which will coincide, by induction).

Induction. Let $M \geq M^\dagger$ and suppose the artificial bounding method is at T_k with the same history as the two-phase procedure. The ratio test (c) and the pivot (d) use only c, A, (B, η) and ρ, which are the same in both methods. It remains to compare (a), (b) and, with (c′), the passing test.

Case T_k in Phase A, so some $\delta_i^{\pm,1} > 0$. Since $\delta_i^{-,1} + \delta_i^{+,1} = l^1_{\beta(i)} - u^1_{\beta(i)} \leq 0$, at most one side of a row has positive slope. In $\mathrm{P}(D_M)$, row i is violated if and only if $\delta_i \succ 0$; rows with $\delta_i^1 > 0$ are violated, rows with $\delta_i^1 < 0$ are not. The maximum of δ_i/ω_i with respect to $\succ$ is attained among rows maximizing $\delta_i^1/\omega_i > 0$, and among them by those maximizing δ_i^0/ω_i, with remaining ties broken by τ_{row}. This is exactly the intercept-refined rule on $\mathrm{P}(D^1)$, whose violated rows are those with $\delta_i^1 > 0$. The side, hence ρ, is the side of positive slope in both. In (c′), all bounds of D_M and D^1 are finite, so the same candidates are eligible for passing. The widths in D_M are $w_j^0 + Mw_j^1$ with $w_j^1 := u_j^1 - l_j^1$, the width in $\mathrm{P}(D^1)$, so the remaining violation in $\mathrm{P}(D_M)$ is $s^0 + Ms$, with s the remaining violation in $\mathrm{P}(D^1)$. A candidate is passed in $\mathrm{P}(D_M)$ if and only if $s^0 + Ms - |\alpha_j|(w_j^0 + Mw_j^1) \succ 0$, which is the intercept-refined passing test.

Case T_k in Phase B. By the invariant and the second property of compatible states, $x^1 = \xi$, which is feasible for the slope problem, so $\delta_i^{\pm,1} \leq 0$ for all i. If $\delta_i^{-,1} < 0$, then $\xi_{\beta(i)} > l^1_{\beta(i)}$, the lower bound of $\beta(i)$ in $\mathrm{P}(D^\xi)$ is $-\infty$, and the side is violated in neither problem. If $\delta_i^{-,1} = 0$, then $\xi_{\beta(i)} = l^1_{\beta(i)}$, $\lambda^\xi_{\beta(i)} = l^0_{\beta(i)}$, and by the third property the constant term $\delta_i^{-,0}$ equals the violation in $\mathrm{P}(D^\xi)$; the same holds for the upper side. Hence the violated rows of $\mathrm{P}(D_M)$ are those of $\mathrm{P}(D^\xi)$, with $\delta_i(M)$ equal to the violation in $\mathrm{P}(D^\xi)$, and (a) and (b) coincide, including the stopping test in (a). In (c′), the remaining violation in $\mathrm{P}(D_M)$ starts with slope $\delta_r^1 = 0$. Let j be a candidate with $\eta(j) = L$ (the case U is symmetric), so $\xi_j = l_j^1$. If $w_j^1 > 0$, passing j would make the slope negative, so j is not passed in $\mathrm{P}(D_M)$; and $\xi_j = l_j^1 < u_j^1$ gives $\mu_j^\xi = +\infty$, so j cannot be passed in $\mathrm{P}(D^\xi)$ either. If $w_j^1 = 0$, then j has two finite bounds in (1), its width is $u_j - l_j$ in both problems, the slope stays 0, and both tests compare the same constant. Variables in status Z are passed in neither.

By induction both methods visit $T_0, \ldots, T_K$ and stop identically at T_K, proving (i) and (ii); (iii) follows from Lemma 1 and the properties of compatible states. □

Termination of both phases holds, for example, under the lexicographic rule for τ_{col} [4], which is compatible with any choice of the leaving row. The threshold $M^\dagger$ depends on the instance and is not known in advance; the point of the theorem is that Phases A and B compute the path of the artificial bounding method for all $M \geq M^\dagger$ without choosing M. The only places where data other than D^1 enter Phase A are the intercept tie-breaks; it reproduces the behavior of the artificial bounding method exactly, but any other tie-breaking rule in Phase A leaves the statements of Section 4 valid. Implementations usually exclude fixed variables from the ratio test and reset their statuses by the signs of their reduced costs after each pivot. If the variables fixed in (1) are treated in this way in all problems, the theorem remains valid with the same proof, since these variables are fixed in every $\mathrm{P}(D)$ and the reset changes no basic solution.

4. Consequences

Corollary 3 (Classification)**.** *Under the assumptions of Theorem 2 (with any tie-breaking rule in Phase A):*

(i) Phase B stops with infeasibility if and only if $F = \emptyset$.

(ii) Phase B stops at an optimal state and $z^1 < 0$ if and only if (1) *is feasible and unbounded.*

(iii) Phase B stops at an optimal state and $z^1 = 0$ if and only if (1) *has a finite optimal value. This value is $c^\top x^0$ for the final basic solution x^0 of Phase B, and $x^0 + M\xi$ is an optimal solution of* (1) *for all sufficiently large M.*

Proof. Since ξ is feasible for the slope problem, $A\xi = 0$ and $l^1 \leq \xi \leq u^1$. By (2): if y is feasible for $\mathrm{P}(D^\xi)$, then $y + M\xi \in F$ for all large M; conversely, if $x \in F$, then $x - M\xi$ is feasible for $\mathrm{P}(D^\xi)$ for all large M. In both cases the objective values differ by Mz^1. Thus $\mathrm{P}(D^\xi)$ is feasible if and only if $F \neq \emptyset$, which gives (i). If $F \neq \emptyset$, then C is the recession cone of F, and since $z^1 = \min\{c^\top d : d \in C \cap [-1, 1]^n\}$, we have $z^1 < 0$ if and only if $c^\top d < 0$ for some d in the recession cone, i.e., if and only if (1) is unbounded [1]. This gives (ii) and the first part of (iii). If $z^1 = 0$, then every $x \in F$ satisfies $c^\top x = c^\top(x - M\xi) \geq c^\top x^0$, and $x^0 + M\xi \in F$ attains this value for large M. □

By Theorem 2, the same classification holds for the artificial bounding method with any $M \geq M^\dagger$: at an optimal state of $\mathrm{P}(D_M)$, the coefficient of M in the objective value, $c^\top x^1$, is negative exactly when (1) is unbounded. Hence the ambiguity of the artificial bounding method between "M too small" and "unbounded" disappears once M is treated through Phases A and B. If $z^1 = 0$ and $\xi \neq 0$, the optimal state of Phase B is optimal for (1) but some nonbasic variables still sit at artificial bounds; they have zero reduced cost (Corollary 5(iii) below). An optimal basis of (1) is then obtained either by the repositioning of Corollary 5(iii) before Phase B, or by removing the coefficient of M after Phase B with at most K pivots that compare affine functions of M lexicographically (Proposition 6).

Remark 4 (Deciding feasibility when $z^1 < 0$). When $z^1 < 0$, problem (1) has no finite optimum, and it remains to decide whether F is empty. Standard options are Phase 1 of the primal simplex method, which minimizes the sum of primal infeasibilities [2, 1], and the dual simplex method on (1) with c replaced by 0, for which every basis is dual feasible. Corollary 3 provides another option: the dual simplex method on the intercept problem, started from the final basis of Phase A, which is dual feasible for it without any change of the objective. If $F \neq \emptyset$, it ends with a point $x^0 + M\xi \in F$ (for large M) and a direction ξ of the recession cone with $c^\top \xi = z^1 < 0$, that is, with a certificate of unboundedness. This option is implemented in the author's open-source solver ENOMOTO-Solver [5]. We do not claim that this option is computationally preferable to the others.

Corollary 5 (Relation to the subproblem approach)**.** *Suppose the subproblem approach solves its auxiliary problem with the same starting state and rules as Phase A. Let $M \geq M^\dagger$.*

(i) Phase 1 of the subproblem approach and the first part of the artificial bounding method visit the same states.

(ii) If $z^1 = 0$ and $\xi = 0$, then the intercept problem is (1)*, and the two methods visit the same states until termination.*

(iii) If $z^1 = 0$ and $\xi \neq 0$, every nonbasic j with $\xi_j \neq 0$ has $\bar{c}_j = 0$. The subproblem approach moves these variables to their finite bound (free variables to status Z), which preserves dual feasibility and makes the slope 0*, and then solves* (1)*. The artificial bounding method keeps them at their artificial bounds and solves* P(D^ξ)*. The two methods differ only by this change of status; after the repositioning, no cleanup (Proposition 6) is needed.*

(iv) If $z^1 < 0$, the subproblem approach stops without distinguishing infeasibility from unboundedness, whereas the artificial bounding method continues with Phase B and decides between them (Corollary 3).

Proof. (i) and the first claim of (ii) are Theorem 2 and $D^\xi = (b, l, u)$ for $\xi = 0$. For (iii), at (B^*, η^*) we have $z^1 = c^\top \xi = \sum_{j \in N} \bar{c}_j \xi_j$, because $\xi_B = -A_B^{-1} A_N \xi_N$. Dual feasibility makes every term nonpositive: $\xi_j \in \{-1, 0\}$ with $\bar{c}_j \geq 0$ for status L, $\xi_j \in \{0, 1\}$ with $\bar{c}_j \leq 0$ for status U, and $\xi_j = 0$ for status Z. Hence $z^1 = 0$ forces $\bar{c}_j = 0$ whenever $\xi_j \neq 0$, so these variables may be placed at any bound without losing dual feasibility. After the move, $\xi_N = 0$ and hence $\xi = 0$. (iv) is Corollary 3. □

Without the repositioning of Corollary 5(iii), when $z^1 = 0$ and $\xi \neq 0$ the final state of Phase B still carries the coefficient $x^1 = \xi$ of M. It can be removed without fixing M, as follows.

Proposition 6 (Removing the coefficient of M)**.** *Suppose Phase B stops at an optimal state with $z^1 = 0$, and let K be the number of nonbasic j with $\xi_j \neq 0$. Starting from this state, repeat the following step while some nonbasic j has $x^1_j \neq 0$, where $x(M) = x^0 + Mx^1$ is the basic solution of the current state for D_M. Choose such a j, and let g_j be the finite bound of j in* (1) *(u_j if $x^1_j = -1$, l_j if $x^1_j = +1$), or $g_j := 0$ if j is free. Move j toward g_j,*

$$x_j \leftarrow x_j(M) - x^1_j\, t, \qquad x_B \leftarrow x_B(M) + t\, a,$$
$$a := x^1_j A_B^{-1} A_j, \qquad 0 \leq t \leq T := |x_j(M) - g_j|.$$

For each row i with $a_i > 0$ and $u_{\beta(i)} < +\infty$ let $t_i := (u_{\beta(i)} - x_{\beta(i)}(M))/a_i$, and for each row with $a_i < 0$ and $l_{\beta(i)} > -\infty$ let $t_i := (x_{\beta(i)}(M) - l_{\beta(i)})/(-a_i)$. These are affine functions of M, and so is T. Let t^* be the minimum of T and the t_i with respect to $\succ$, ties broken in favor of T and then by the smallest index.

(A) If $t^* = T$, place j at g_j (status U, L or Z, respectively); the basis is unchanged.

(B) If $t^* = t_r$, pivot j into the basis in row r; $\beta(r)$ becomes nonbasic at the finite bound it has reached.

The procedure stops after exactly K steps at an optimal basis of (1)*, i.e., a dual feasible state admissible for (b, l, u) whose basic solution is feasible for* (1)*, and its objective value is $c^\top x^0$ of the final state of Phase B.*

Proof. At the start, the state is dual feasible, $\bar{c}_j = 0$ whenever $x^1_j = \xi_j \neq 0$ (Corollary 5(iii)), and $x(M) \in F$ for all large M (proof of Corollary 3). We show that each step preserves these properties and the objective value, and decreases the number of nonbasic j with $x^1_j \neq 0$ by one. The objective changes by $-x^1_j t\, \bar{c}_j = 0$. In (A) the basis is unchanged and only the status of j changes, which is admissible since $\bar{c}_j = 0$. In (B) the pivot column has $\bar{c}_j = 0$, so the reduced costs of the other nonbasic variables do not change and that of $\beta(r)$ becomes 0; dual feasibility is kept, and in particular $\bar{c}_k = 0$ still holds for the remaining k with $x^1_k \neq 0$. Since $x(M)$ satisfies the finite bounds of (1) for large M, every $t_i \geq 0$, and T has slope 1; hence $0 \leq t^* \leq T$, the moving variable stays between $x_j(M)$ and g_j, and no basic variable passes a finite bound. The bounds that are infinite in (1) impose nothing. Thus the new basic solution, again affine in M, lies in F for large M. Finally, j becomes basic or is placed at g_j with slope 0, $\beta(r)$ is placed at a finite bound with slope 0, and no other status changes. After K steps $x^1_N = 0$, hence $x^1 = -A_B^{-1} A_N x^1_N = 0$, and the basic solution is a constant vector in F. All nonbasic variables are at finite bounds of (1) or in status Z, so the state is an optimal basis of (1). □

This is the primal push of crossover [16, 15], carried out on affine functions of M. It is the only place where both components of $\succ$ are needed: T and some t_i may have the same slope, or two t_i may. In contrast to substituting a numerical M into $x^0 + M\xi$, which gives an optimal solution that is in general not basic and may require a very large M, the procedure returns a basic solution and never fixes M.

Example 7. Consider $\min x_1$ subject to $x_2 + s = 5$, x_1 free, $0 \leq x_2 \leq 1$, $s \in [0, 0]$, started from the slack basis with x_1 at status L (since $c_1 = 1 > 0$) and x_2 at L. In the slope problem, $x_1 = -1$ and $s = 0$, so Phase A stops at once with $\xi = (-1, 0, 0)$ and $z^1 = -1$; the subproblem approach reports dual infeasibility. The intercept problem has $x_1 \geq 0$, $0 \leq x_2 \leq 1$, $s \in [0, 0]$, and Phase B detects $E = \emptyset$ after one pivot, so the problem is infeasible (not unbounded). The artificial bounding method with $M \geq M^\dagger$ performs the same pivot. Note that its first violation, $\delta(M) = 5$, has slope 0; a rule that compares slopes only would stop with $z^1 < 0$ and misreport unboundedness.

Remark 8 (Dual interpretation). Writing $\kappa, \nu \geq 0$ for the multipliers of the lower and upper bounds, the dual of P(D_M) is

$$\begin{aligned} \max\ & b^\top y + \textstyle\sum_{l_j > -\infty} l_j \kappa_j - \sum_{u_j < +\infty} u_j \nu_j \\ & - M\big(\textstyle\sum_{l_j = -\infty} \kappa_j + \sum_{u_j = +\infty} \nu_j\big) \\ \text{s.t. } & A^\top y + \kappa - \nu = c, \quad \kappa, \nu \geq 0. \end{aligned}$$

The dual of (1) is obtained by fixing the multipliers of the infinite bounds at 0 ($\kappa_j = 0$ if $l_j = -\infty$, $\nu_j = 0$ if $u_j = +\infty$). Hence the dual of P(D_M) allows these multipliers as artificial variables with penalty M, i.e., it is the big-M method for

the dual [9]. Since the right-hand side and the finite bounds of the slope problem are 0, its dual has only the objective $-(\sum_{l_j=-\infty} \kappa_j + \sum_{u_j=+\infty} \nu_j)$, i.e., it minimizes the sum of the artificial variables (dual Phase 1). Since the dual simplex method on the primal is the primal simplex method on the dual [12], Theorem 2 states that the big-M method on the dual follows, pivot by pivot, the two-phase method on the dual. This is an iteration-level counterpart of the classical limit statement [2, 1], obtained without non-Archimedean arithmetic [3]: in the main iterations the two components of the lexicographic comparison are never active at the same time. Proposition 6 corresponds on the dual side to driving the artificial variables that remain basic at value 0 out of the basis.

5. Conclusion

The artificial bounding method, run with any sufficiently large M, is the subproblem approach followed by the dual simplex method on an intercept problem, and the two Phase 1 methods differ only in the placement of zero-reduced-cost variables at the end of Phase 1. The practical drawback of the artificial bounding method, the choice of M, is therefore not intrinsic: Phases A and B realize its limit behavior with ordinary dual simplex iterations on problems with real data, the intercept phase settles the case of dual infeasibility, which the subproblem approach leaves open, and a lexicographic primal push removes the coefficient of M that may remain at the end.

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